WORKSHEET

Chapter 8: Motion with variable forces

Use $g = 9.8 \text{ m s}^{-2}$ unless otherwise instructed.

1 In each of the following situations, m = mass (kg), u = initial velocity (m s^{-1}) and F = force (N). Use this information to find an expression for velocity as a function of time, $v(t)$.

a $m = 1$, $u = 0$ and $F = t - 2t^2$

b $m = 5$, $u = 0$ and $F = \sin(2t + 1)$

c $m = 2$, $u = 3$ and $F = 3e^{0.4t + 5}$

d $m = 6$, $u = 4$ and $F = 4v$

e $m = 3$, $u = 2$ and $F = v + 5$

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2 A particle of mass 3 units moves in a straight line under the action of a force of $F = 6\cos(t)$. At time t, the displacement of the particle is x and the speed is v. If $v = 4$ and $x = 0$ when $t = 0$, find x in terms of t.

3 A particle of mass 3 units moves in a straight line under the action of a force of $F = 4 + 8x$. At time t, the displacement of the particle is x and the speed is v. If $v = 4$ and $x = 0$ when $t = 0$, find v when $x = 1$.

4 An object of mass 5 kg moves from rest under the action of a force with magnitude after t s given by $t(t + 1)$ N. Calculate the speed of the object after 3 seconds and the distance travelled.

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5 A force of $4x + 5$ acts on a particle of mass 2 kg moving in a straight line where x is the displacement from a fixed origin. The initial velocity, v, is 4 m s^{-1} when $x = 0$. Find the velocity when $x = 1$.

6 An object of mass 4 kg is initially at rest. The object is acted on by a force of $(6 - t)^2$ N where $0 \leq t \leq 6$. Find the speed of the object after 6 seconds and the distance travelled.

7 An object of mass 4 kg is acted on by a force (measured in newtons) whose magnitude is given by $\mathbf{F} = 4(3e^{-0.1t} + 2t)$.

If the object has an initial velocity of 8 m s^{-1}, find

a the velocity when $t = 10$

b the displacement in the first 2 seconds of its motion.

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8 A particle of mass m falls under gravity from rest through the air. Air resistance, $\mathbf{r} = k\mathbf{v}$. Find an expression for:

a velocity as a function of time

b terminal velocity (Hint: Terminal velocity is reached as $t \to \infty$)

c displacement as a function of time.

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Answers

1 **a** $v(t) = t^2 - \frac{2t^3}{3}$

b $v(t) = \frac{-1}{10}\cos(2t + 1)$

c $v(t) = 10e^{0.2t + 3} - 195.86$

d $v(t) = 4e^{\frac{-2t}{3}}$

e $v(t) = 7e^{\frac{-t}{3}} - 5$

2 $x = 2 - 2\cos(t) + 4t$

3 $v = \pm \frac{2\sqrt{30}}{3}$

4 $2\frac{7}{10}$ m s^{-1}, $2\frac{1}{4}$ m

5 $v = \pm 5$ m s^{-1}

6 18 m s^{-1}; 72 m

7 **a** About 127 m s^{-1}

b About 24.3 m

8 **a** $v = \frac{g}{k}(1 - e^{-kt})$

b $\frac{g}{k}$

c $x = \frac{gt}{k} - \frac{g}{k^2}(1 - e^{-kt})$

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